SHIV NADAR UNIVERSITY

Programme: **B. Tech** Discipline: **Computer Science and Engineering**
Exam: **End Semester** Year: **2019 - 2020**
Course Code: **CSD301** Course Title: **Software Engineering**
Date: **December 03, 2019** Time: **9:30 AM – 12:30 PM** Max. Marks: **80**

Ques.1 Choose the Single Correct Option:- **(1*5 = 5 Marks)**

(i) The radial dimension of the spiral model represents which one of the following:

(a) Cost incurred so far
(b) Progress in the implementation of the current feature
(c) Number of features implemented so far
(d) Number of risks that have been resolved so far

(ii) Which one of the following is the correct order in which a software project manager estimates various project parameters while using the COCOMO estimation technique?

(a) Cost, effort, duration, size
(b) Size, effort, duration, cost
(c) Cost, duration, effort, size
(d) Size, cost, effort, duration

(iii) The PERT method differs from CPM in which one of the following aspects:

(a) PERT uses statistical time durations, whereas CPM uses deterministic time durations
(b) PERT uses dummy activities, whereas CPM does not
(c) PERT uses free float, whereas CPM uses total float in critical path calculations
(d) PERT uses activity on arc, whereas CPM uses activity on node networks

(iv) Which of the following statements is not true of Data Flow Diagrams (DFDs)?

(a) Hierarchical Diagram
(b) Do not represent decisions and control flows
(c) Represent code structure
(d) Represent functional decomposition

(v) Consider the following partial requirement for a software: "A web browser should be used as the front end." It can be considered to be which one of the following types of requirements:

(a) Functional requirement **(b)** Non-functional requirement **(c)** Constraint **(d)** External Interface

Ques.2 For a program with number of unique operators, $\eta_1 = 20$ and number of unique operands $\eta_2 = 40$, compute the following. Also, $N_1 = 25$ and $N_2 = 50$. **(3 Marks)**

(a) Program Volume **(b)** Program Length **(c)** Effort and Time

Ques.3 Answer the following questions in not more than 40-50 words: **(2*8 = 16 Marks)**

(a) What do you mean by balancing a DFD? Illustrate your answer with suitable example.

(b) List two deficiencies in Waterfall Model. Which process model do you suggest to overcome each deficiency?

(c) Mention 4 major drawbacks of Spiral Model.

(d) Define Data Dictionary with an example.

(e) What are the issues in measuring the software size using LOC as metric?

(f) What is Cyclomatic complexity?

(g) Draw the Structure Chart for the RMS software.

(h) Using the data of the following table, what can be said about the total project duration?

Activity	Duration	Predecessors
A	14	-
B	13	A
C	20	A
D	10	B, C

Ques.4 Assume that you are the technical manager of a software development organization. A client approached you for a software solution. The problems stated by the client have uncertainties which lead to loss if it not planned and solved. Which software development model you will suggest for this project-justify. Explain that model with its pros and cons and neat sketch. **(5 Marks)**

Ques.5 Identify the causes and effects and draw a cause-effect graph for the following situation of a bank database. Suppose only two commands are allowed: **(5 Marks)**

1. Credit account number transaction amount
2. Debit account number transaction amount

The requirements are: if the command is credit and the account number is valid, then account is credited. If the command is debit, the account number is valid and the transaction amount is valid (less than the balance), then account is debited. If the command is not valid, the account number is not valid or the debit amount is not valid, a suitable message is generated.

Ques.6 Define Coupling. Also explain its different types. **(5 Marks)**

Ques.7 Draw the **sequence diagram** for the "**Renew Book**" use case of a Library Management System Scenario. **(5 Marks)**

Ques.8 What are the various requirements to achieve MC/DC based structural testing. Identify the test case suites using decision table for the expression: **(6 Marks)**

1) if((A && B) && (C || D))

Ques.9 The BSE Electrical Company charges its domestic consumers using the following Slab: **(10 Marks)**

Consumption units	Energy charges
0-150	2.00 per unit
151-300	Rs 200 + Rs. 3.00 per unit in excess of 150 units
301-400	Rs. 300 + Rs 3.90 per unit in excess of 300 units
>400	Rs. 350 + Rs 4.40 per unit in excess of 400 units

(i) Write a C program that prints the energy charges according to the given slab.
(ii) Draw a Control flow graph of the C program you generated for the BSE.
(iii) Compute the cyclomatic complexity of the control flow graph and identify the basis paths.
(iv) Identify the equivalence classes and equivalence class test cases for input domain along with the charges incurred.
(v) Identify the robust test cases for input domain and the charges incurred

Ques.10 Construct DFD for the following system. **(10 Marks)**

An online shopping system for XYZ provides many services and benefits to its members and staffs. Currently, XYZ staffs manually handle the purchasing information with the use of basic office software, such as Microsoft Office Word and Excel. It may results in having mistakes easily and the process is very inconvenient. XYZ needs an online shopping system at their Intranet based on the requirements of users. XYZ online shopping system has five key features:

i) To provide the user friendly online shopping cart function to members to replace hardcopy ordering form;
ii) To store inventory and sales information in database to reduce the human mistakes, increase accuracy and enhance the flexibility of information processing;
iii) To provide an efficient inventory system which can help the XYZ staffs to gain enough information to update the inventory;
iv) To be able to print invoices to members and print a set of summary reports for XYZ's internal usage;
v) To design the system that is easy to maintain and upgrade

Ques.11 The time estimates (in hours) for the activities of a PERT network are given below: **(10 Marks)**

Activity	T_O	T_M	T_P
1-2	1	1	7
1-3	1	4	7
1-4	2	2	8
2-5	1	1	1
3-5	2	5	14
4-6	2	5	8
5-6	3	6	15

where, T_O is the optimistic time, T_P is the pessimistic time and T_M is the most likely time

(i) Draw the project network
(ii) Identify all paths through it and write critical path
(iii) Determine the expected project length

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